جامعة الشارقة
UNIVERSITY OF SHARJAH

College of Sciences

Department of Computer Science

Course Syllabus

Course Title

Web Programming

Course Code

1501341

Fall Semester, 2022 / 2023

<u>Course Syllabus</u>

Course Title	Web Programming
Course Code	1501341
Credit Hours	3
Pre-requisite(s)	1411116 Programming I
Co-requisite(s)	None
Semester	Fall	**Year**	2022/2023
Instructors Name	Dr. Isam Al Jawarneh
Office Location	M5 - 211
Tel. No.	+(971) 6 5052591
Email	ijawarneh@sharjah.ac.ae
Lecture Times	TR 11:00-12:15 (section 12)
Office Hours	Monday 12:30 – 13:30 & Thursday 12:30 – 13:30 & By appointment

Course Description (as in the catalogue):

Introduction to **H**yper**T**ext **M**arkup **L**anguage (HTML5): Tags, headers, text style, fonts, line breaks, rules, linking, images, lists, tables, forms, and frames. Semantic tags, Canvas, Geolocation, JQuery, Drag and Drop. Dynamic HTML: Cascading Style Sheets: Inline styles, external style sheets, backgrounds, positioning elements, text flow and box model. Filters: Flip, grayscale, sepia, saturate, hue-rotate, invert, opacity, blur, brightness, contrast, drop-shadow. JavaScript: A simple program, memory concepts, assignment operators, decision making, control structures, if-else, while, repetition, for, switch, do/while, functions, arrays. Object Model and Collections: all, children. Event Model: *OnClick, OnLoad, OnError, OnMouseMove, OnMouseOver, OnMouseOut, OnFocus, OnBlur, OnSubmit, OnReset*. Multimedia. DHTMLMenu builder. PHP and databases.

Course Objectives/Goals:

The goals of this course are to:

- Introduce features of the Internet and the WWW.
- Provide practical experience for web-design techniques and tools.
- Appreciate the need and use of scripting techniques.
- Introduce multimedia tools and technologies.

Course Learning Outcomes:

By the end of successful completion of this course, the student will be able to:

1. HTML to design, implement and administrate web-sites using HTML.
2. Make use of scripting techniques to create dynamic parts of web pages.
3. Add special effects and use event models to enhance web pages.
4. Acquire practical knowledge of some contemporary software packages for web-design.
5. Use multimedia technologies to improve web page look and interaction.
6. Participate in a team to design and implement a large web-site.

<u>Major topics:</u>

- An introduction to the Internet & WWW
- Webpages construction: content [HTML]
- Websites presentation: CSS
- Websites behavior (client-side scripting): JavaScript
- Server-side scripting: PHP & form processing
- Data-based interactive web development: Integrating PHP and MySQL.

Alignment of Course Student Learning Outcomes to the BSCS Program Student Learning Outcomes

Computer Science Program SLOs		Course SLOs
Analyze a complex computing problem and to apply principles of computing and other relevant disciplines to identify solutions.	**Analyze**	
	Apply	- Make use of scripting techniques to create dynamic parts of web pages. - Add special effects and use event models to enhance web pages.
Design, implement, and evaluate a computing-based solution to meet a given set of computing requirements in the context of the program’s discipline.	**Design**	- HTML to design, implement and administrate web-sites using HTML. - Add special effects and use event models to enhance web pages. - Use multimedia technologies to improve web page look and interaction.
	Implement	- HTML to design, implement and administrate web-sites using HTML. - Add special effects and use event models to enhance web pages. - Acquire practical knowledge of

Computer Science Program SLOs		Course SLOs
		some contemporary software packages for web-design. - Use multimedia technologies to improve web page look and interaction.
	Evaluate	
Communicate effectively in a variety of professional contexts.		- Participate in a team to design and implement a large web-site.
Describe professional responsibilities and make informed judgments in computing practice based on legal and ethical principles.		
Function effectively as a member or leader of a team engaged in activities appropriate to the program's discipline.		- Participate in a team to design and implement a large web-site.
Apply computer science theory and software development fundamentals to produce computing-based solutions.	**Theory**	
	Development	- HTML to design, implement and administrate web-sites using HTML. - Make use of scripting techniques to create dynamic parts of web pages. - Add special effects and use event models to enhance web pages. - Acquire practical knowledge of some contemporary software packages for web-design. - Use multimedia technologies to improve web page look and interaction.

Alignment of Course Student Learning Outcomes to the BSIT Program Student Learning Outcomes

IT Multimedia Program SLOs		Course SLOs
Analyze a complex computing problem and to apply principles of computing and other relevant disciplines to identify solutions.	**Analyze**	
	Apply	- Make use of scripting techniques to create dynamic parts of web pages. - Add special effects and use event models to enhance web pages.
Design, implement, and evaluate a computing-based solution to meet a given set of computing requirements in	**Design**	- HTML to design, implement and administrate web-sites using HTML.

IT Multimedia Program SLOs		Course SLOs
the context of the program's discipline.		- Add special effects and use event models to enhance web pages. - Use multimedia technologies to improve web page look and interaction.
	Implement	- HTML to design, implement and administrate web-sites using HTML. - Add special effects and use event models to enhance web pages. - Use multimedia technologies to improve web page look and interaction.
	Evaluate	
Communicate effectively in a variety of professional contexts.		- Participate in a team to design and implement a large web-site.
Describe professional responsibilities and make informed judgments in computing practice based on legal and ethical principles.		
Function effectively as a member or leader of a team engaged in activities appropriate to the program's discipline.		- Participate in a team to design and implement a large web-site.
Identify and analyze user needs and to take them into account in the selection, creation, integration, evaluation, and administration of computing-based systems.	**Identify / Analyze**	
	Development	- HTML to design, implement and administrate web-sites using HTML. - Make use of scripting techniques to create dynamic parts of web pages. - Add special effects and use event models to enhance web pages. - Use multimedia technologies to improve web page look and interaction.

Weekly Distribution of Course Topics/Contents

Week	Topic	Book chapter	Assignments	Course SLO
1 & 2	Course overview Introduction: Internet & Web	Ch1 (in addition to Multiple other Resources)		1,4
3 & 4	**Content structure**: introduction to **HTML**	Ch2		1, 4
5	Content structure: Advanced **HTML**	Ch3	HW#1	1,5
5	Content **presentation**: introduction to **CSS**	Ch4		3
6	Content **presentation**: Advanced **CSS**	Ch5	HW#2	3,5,6
7	Introduction to **Client-side scripting**: **JavaScript** introduction	Ch6		2,6
8	Introduction to **Client-side scripting**: **JavaScript** control statements	Ch7 & 8		2,6
9	Advanced **JavaScript**: functions	Ch9		2,6
10	Advanced **JavaScript**: arrays	Ch10	HW#3	2,6
11	**Client-side** Event Handling	ch13		2, 3,6
12	**Server-side** data storage: Introduction to **MySQL**	Ch18		2, 3,4,6
13	**Server-side** scripting: introduction to **PHP**	Ch19	HW#4	2, 3, 4,6
14	Integrating **PHP** and **MySQL** for **Client-Server** Database Interaction	Ch18 & Ch19		2, 4,6
15	**Revision and Final Exam**			

Students' Assessment:

Students are assessed as follows:

Assessment Tool(s)**	Date	Weight (%)
Project	Week #6 & #14	20
Theory Quizzes (Average of best 3)	1. Week #4 or #5 2. Week #7 or #8 3. Week #10 4. Week #12	10
Homework (Average of 4 homework)	1. Posting Date: Week #5, Due Date: Week #6 (**HTML**) 2. Posting Date: Week #6, Due Date: Week #8 (**CSS**) 3. Posting Date: Week #10, Due Date: Week #12 (Introduction to **JavaScript**) 4. Posting Date: Week #13, Due Date: Week #14. (**Server-side & data storage**)	10
Midterm Exam	Midterm Exam Period	20
Final Theory Exam	Final Exam Period	40
Total		100%

Course Outcome Assessment Plan:

Course LOs	Teaching/Learning Method(s)	Assessment Tool(s)	Performance Indicators
HTML to design, implement and administrate web-sites using HTML.	On-campus (face-to-face) & Live Online Lectures / Recordings Case Studies Team Discussion	- Exams - Assignments - Quizzes - Project	Course assessment Marks Project Demo
Make use of scripting techniques to create dynamic parts of web pages.	On-campus (face-to-face) & Live Online Lectures / Recordings Problem Solving Case Studies Team Discussion	- Exams - Assignments - Quizzes - Project	Course assessment Marks Project Demo Lab Demo
Add special effects and use event models to enhance web pages.	On-campus (face-to-face) & Live Online	- Exams - Assignments	Course assessment

Course LOs	Teaching/Learning Method(s)	Assessment Tool(s)	Performance Indicators
	Lectures / Recordings Labs Case Studies Team Discussion	- Quizzes - Project	Marks Project Demo
Acquire practical knowledge of some contemporary software packages for web-design.	On-campus (face-to-face) & Live Online Lectures / Recordings Team Discussion	- Exams - Assignments - Quizzes - Project	Course assessment Marks Project Demo
Use multimedia technologies to improve web page look and interaction.	On-campus (face-to-face) & Live Online Lectures / Recordings Team Discussion	- Exams - Assignments - Quizzes - Project	Course assessment Marks Project Demo
Participate in a team to design and implement a large web-site.	On-campus (face-to-face) & practical sessions Case Studies Team Discussion	- Exams - Assignments - Quizzes - Project	Course assessment Marks Project Demo

Teaching and Learning Resources:

Required Textbook(s):

- *Internet & World Wide Web: How to Program, Fifth Edition,* Deitel, H. M., Deitel, P. J., Deitel & Associates, Inc. Published by Pearson, Copyright © 2012. eText ISBN-13: **9780137618279.** Paperback ISBN-13: **9780132151009**

References:

- *Web Programming with HTML5, CSS, and JavaScript*, First Edition. John Dean. Jones & Bartlett learning. © 2019. ISBN: 9781284091793
- *Web Programming and Internet Technologies*, 2nd Edition. Porter Scobey. Jones & Bartlett Learning. 2016. ISBN: 9781284070699
- https://www.w3schools.com/

Software

We will use the following tools:

- https://jsfiddle.net/ for testing HTML, CSS & JavaScript online interactively
- Also, a local text editor to run the examples locally on your machine. In addition to a web browser (MS Edge, Google Chrome, Firefox, etc.,)
- The XAMPP integrated installer for Apache Web Server, MySQL and PHP (available for Windows, Mac OS X & Linux)
    - https://www.apachefriends.org/

- GitHub pages for hosting static webpages

Your instructor will supply relevant instructions for using those tools

Attendance policy:

Attendance is compulsory. A student missing 10% (6 hours) of the total allocated course hours will receive 1st warning notice and a student missing 15% (9 hours) will receive 2nd warning notice. A student missing 20% (12 hours or more) will be forced to withdraw (in accordance with the university regulations).

Plagiarism/Cheating:

Students are expected to do their own work. You are allowed to work on assignments in teams only if specified by the instructor. In other words, students are encouraged to communicate about general principles of the course, but all assigned homework must be done on an individual basis. The instructor is available to provide any assistance that you may need. Cheating is considered a serious offense by the university. You should be aware of the severe penalty for cheating (refer to the student code of conduct published in the university catalogue).

Notes:

Practical Aspect of the course:

Students are expected to attend weekly practical classes to practice main concepts of the course. Students are given programming assignments too. Each assignment is usually made of a number of problems.

In-class Activities:

Students are expected to do their own work individually and/or in teams. Your work individually and in teams will be graded. Activities include quizzes, individual assignments, and team assignments.

Hybrid Teaching and Learning:

In Fall 2022/2023, all the lectures will be delivered using hybrid learning. Blackboard Collaborate Ultra will be used to deliver the online lectures. The recordings of the lectures will be made available to the students automatically through the blackboard. All course material will be distributed online using the blackboard. Mid Term and Final Exams will follow the University's regulations with regards to the online or in-person exams. Additional in-class assessment will be conducted online. All submissions from the students for these in-class assessment, or homework assignments will be online using the blackboard. Students will be provided with as many online resources as possible to facilitate their learning.